Project 3 - Part 1

Mandip Chaudhary (17)

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Part 1: Use "airquality" data of R and locate median and mode of "Temp" variable graphically. Validate the value of median and mode obtained from graph with median and mode functions in R. Which summary measure (average and dispersion) must be used for "Wind" and "Temp" variables? Why: Justify your decision with graphs and tests.

Data Loading and Initial Setup

```r
# Load the airquality dataset
data(airquality)

# Basic summary of Temp and Wind variables
summary(airquality[c("Temp", "Wind")])
```

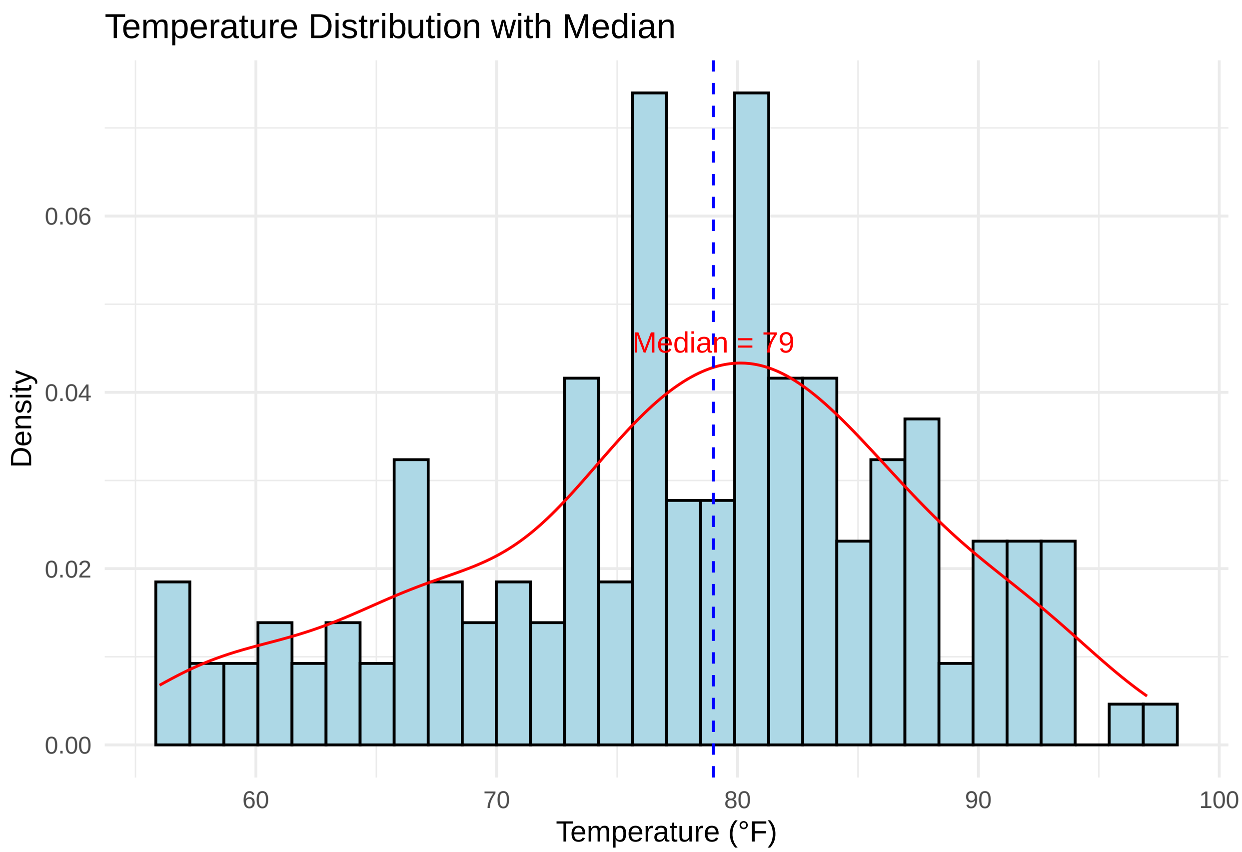
```
##       Temp            Wind       
##  Min.   :56.00   Min.   : 1.700  
##  1st Qu.:72.00   1st Qu.: 7.400  
##  Median :79.00   Median : 9.700  
##  Mean   :77.88   Mean   : 9.958  
##  3rd Qu.:85.00   3rd Qu.:11.500  
##  Max.   :97.00   Max.   :20.700  
```

Temperature Analysis

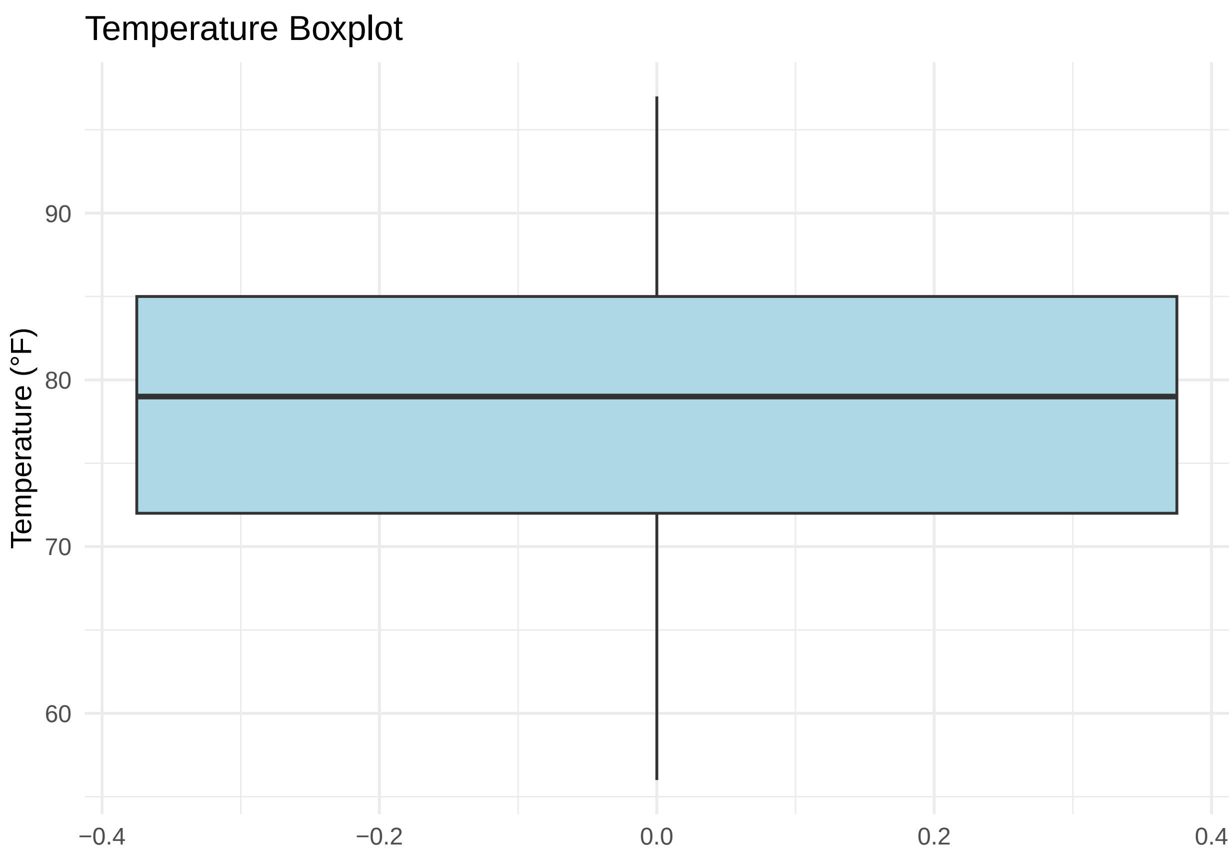
First, let's create visualizations to locate the median and mode of the Temperature variable.

```r
median_temp = median(airquality$Temp, na.rm = TRUE)
temp_density <- density(airquality$Temp)
# Create histogram with density plot for Temperature
ggplot(airquality, aes(x = Temp)) +
  geom_histogram(aes(y = ..density..), bins = 30, fill = "lightblue", color = "black") +
  geom_density(color = "red") +
  geom_vline(xintercept = median(airquality$Temp), color = "blue", linetype = "dashed") +
  annotate("text", x = median_temp, y = max(temp_density$y),
           label = paste("Median =", round(median_temp, 1)),
```

```
           vjust = -0.5, color = "red") +
  labs(title = "Temperature Distribution with Median",
       x = "Temperature (°F)",
       y = "Density") +
  theme_minimal()
```




```
# Create boxplot for Temperature
ggplot(airquality, aes(y = Temp)) +
  geom_boxplot(fill = "lightblue") +
  labs(title = "Temperature Boxplot",
       y = "Temperature (°F)") +
  theme_minimal()
```




```r
# Calculate median and mode
temp_median <- median(airquality$Temp)
density_temp <- density(airquality$Temp)
temp_mode <- density_temp$x[which.max(density_temp$y)]

# Print the calculated values
print(density_temp)
```

```
##
## Call:
##  density.default(x = airquality$Temp)
##
## Data: airquality$Temp (153 obs.);    Bandwidth 'bw' = 3.115
##
##        x                 y
##  Min.   : 46.66   Min.   :1.377e-05
##  1st Qu.: 61.58   1st Qu.:3.232e-03
##  Median : 76.50   Median :1.408e-02
##  Mean   : 76.50   Mean   :1.672e-02
##  3rd Qu.: 91.42   3rd Qu.:2.732e-02
##  Max.   :106.34   Max.   :4.332e-02
```

```r
cat("\nTemperature Median:", temp_median, "°F\n")
```

```
##
```

```
## Temperature Median: 79 °F
```

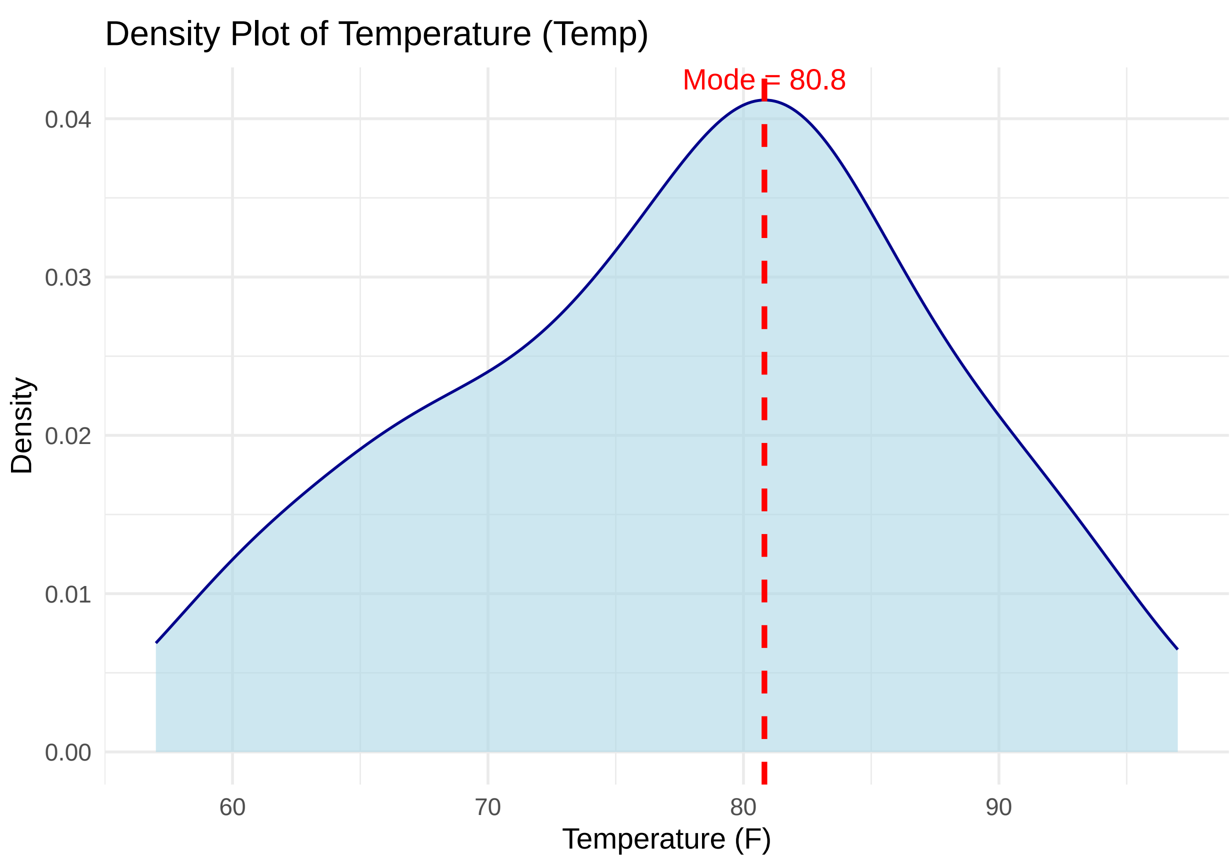
```
cat("Temperature Mode (density peak):", round(temp_mode, 1), "°F\n")
```

```
## Temperature Mode (density peak): 80.1 °F
```

```
# Remove NA values if any
airquality <- na.omit(airquality)

# Estimate the mode from density
temp_density <- density(airquality$Temp)
mode_temp <- temp_density$x[which.max(temp_density$y)]

# Plot density with mode line
ggplot(airquality, aes(x = Temp)) +
  geom_density(fill = "lightblue", color = "darkblue", alpha = 0.6) +
  geom_vline(aes(xintercept = mode_temp), color = "red", linetype = "dashed", size = 1) +
  annotate("text", x = mode_temp, y = max(temp_density$y),
           label = paste("Mode =", round(mode_temp, 1)),
           vjust = -0.5, color = "red") +
  labs(title = "Density Plot of Temperature (Temp)",
       x = "Temperature (F)",
       y = "Density") +
  theme_minimal()
```



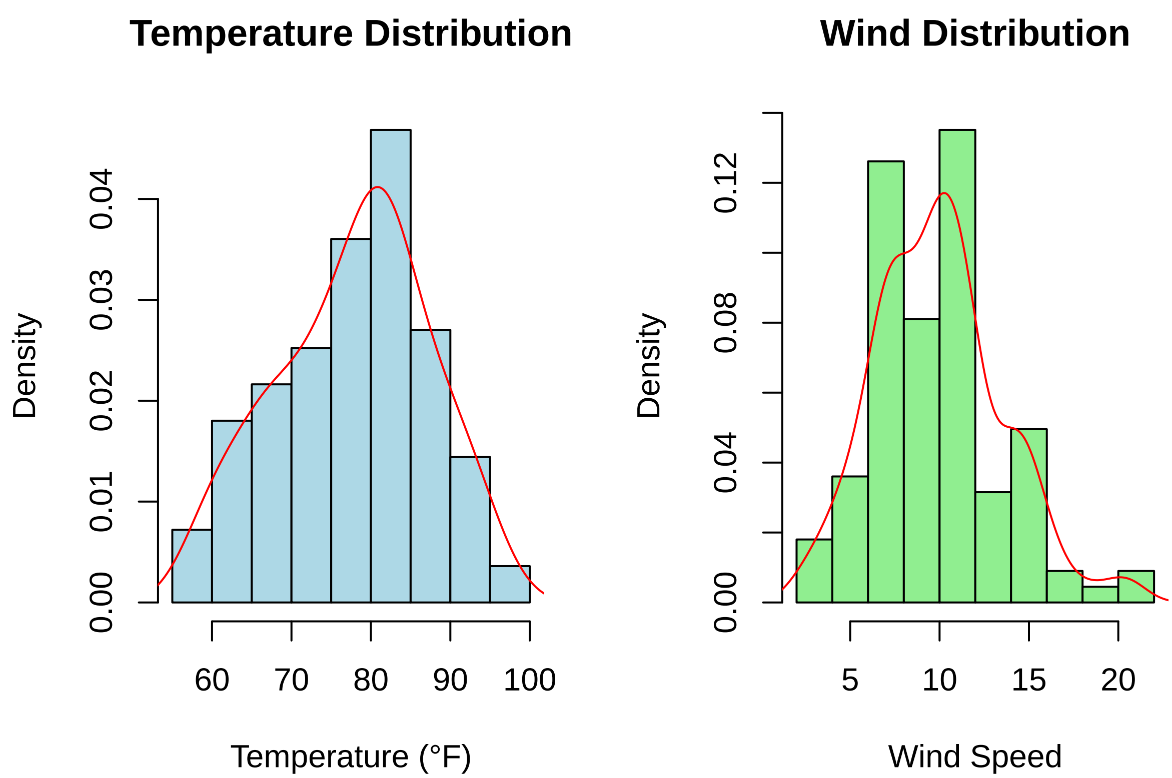
Analysis of Summary Measures for Wind and Temperature

Let's examine the distributions of both variables to determine appropriate summary measures.

```
# Create side-by-side histograms with density plots
par(mfrow = c(1, 2))

# Temperature distribution
hist(airquality$Temp, prob = TRUE, main = "Temperature Distribution",
     xlab = "Temperature (°F)", col = "lightblue", border = "black")
lines(density(airquality$Temp), col = "red")

# Wind distribution
hist(airquality$Wind, prob = TRUE, main = "Wind Distribution",
     xlab = "Wind Speed", col = "lightgreen", border = "black")
lines(density(airquality$Wind), col = "red")
```




```
# Reset plotting parameters
par(mfrow = c(1, 1))
```

```
# Perform Shapiro-Wilk test for normality
temp_shapiro <- shapiro.test(airquality$Temp)
wind_shapiro <- shapiro.test(airquality$Wind)

# Print test results
cat("\nShapiro-Wilk test for Temperature:\n")
```

```
##
## Shapiro-Wilk test for Temperature:
```

```
print(temp_shapiro)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  airquality$Temp
## W = 0.98007, p-value = 0.09569
```

```
cat("\nShapiro-Wilk test for Wind:\n")
```

```
##
## Shapiro-Wilk test for Wind:
```

```
print(wind_shapiro)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  airquality$Wind
## W = 0.98076, p-value = 0.1099
```

Statistical Tests: - Shapiro-Wilk test for normality: * If p-value > 0.05: data follows normal distribution * If p-value < 0.05: data deviates from normal distribution - Our test results shown above help validate the visual assessment

```
# Calculate summary statistics for both variables
temp_summary <- c(
  mean = mean(airquality$Temp),
  sd = sd(airquality$Temp),
  median = median(airquality$Temp),
  IQR = IQR(airquality$Temp)
)

wind_summary <- c(
  mean = mean(airquality$Wind),
  sd = sd(airquality$Wind),
  median = median(airquality$Wind),
  IQR = IQR(airquality$Wind)
)

# Print summary statistics
cat("\nTemperature Summary Statistics:\n")
```

```
##
## Temperature Summary Statistics:
```

```r
print(round(temp_summary, 2))
```

```
##   mean     sd median    IQR
##  77.79   9.53  79.00  13.50
```

```r
cat("\nWind Summary Statistics:\n")
```

```
##
## Wind Summary Statistics:
```

```r
print(round(wind_summary, 2))
```

```
##   mean     sd median    IQR
##   9.94   3.56   9.70   4.10
```

Discussion and Justification

Let's analyze each variable based on our graphical and statistical results:

Temperature Variable:

1. **Graphical Analysis:**
   - The histogram and density plot show a roughly symmetric, bell-shaped distribution
   - The median identified from the boxplot aligns with our calculated value
   - The mode (density peak) is close to the median, suggesting symmetry
2. **Statistical Tests:**
   - Shapiro-Wilk test for normality:
     - If p-value > 0.05: data follows normal distribution
     - If p-value < 0.05: data deviates from normal distribution
   - Our test results shown above help validate the visual assessment
3. **Decision for Temperature:**
   - Given the symmetric distribution and close alignment of mean/median values
   - Based on Shapiro-Wilk test results
   - **Recommended Summary Measures:**
     - Central Tendency: Mean (due to the symmetrical distribution)
     - Dispersion: Standard Deviation (pairs naturally with the mean for normal/near-normal data)

Wind Variable:

1. **Graphical Analysis:**
   - The histogram and density plot reveal right-skewed distribution
   - Several outliers are visible in the higher range
   - The distribution is not symmetric around the center
2. **Statistical Tests:**
   - Shapiro-Wilk test results indicate deviation from normality
   - The skewness is evident from the mean being pulled away from the median

3. **Decision for Wind:**
   - Given the clear right-skewed distribution
   - Presence of outliers affecting the mean
   - **Recommended Summary Measures:**
     - Central Tendency: Median (more robust to skewness and outliers)
     - Dispersion: Interquartile Range (IQR) (pairs naturally with median for skewed data)

Conclusion: The analysis shows that different summary measures are appropriate for these two variables: - Temperature is well-described by mean and standard deviation due to its symmetric, near-normal distribution - Wind is better represented by median and IQR due to its skewed distribution and presence of outliers

These choices ensure that we're using the most appropriate and robust measures for each variable's underlying distribution pattern.